

CPTG: 123/50 Suppression Factor and Early Galaxy Formation

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Abstract

In Curvature Polarization Transport Gravity, the suppression factor prevents the nonlinear curvature response from becoming unstable. The exponent 123/50 acts as a governor on curvature polarization, especially in dense early-universe environments. This allows rapid galaxy formation while preventing runaway collapse or unphysical luminosity growth.

1 Suppression Factor

The CPTG suppression factor is

$$\Sigma = \left[1 + \left(\frac{g}{a_{\star}} \right)^{123/50} \right]^{-11/50}.$$

Here, the local acceleration field is

$$g,$$

and the CPTG acceleration scale is

$$a_{\star}.$$

The exponent is

$$\frac{123}{50} = 2.46.$$

This exponent controls how sharply the nonlinear response is suppressed when the local field becomes strong.

2 Saturation Limit

In the early universe, matter density is much higher than it is in the present epoch. This means local gravitational acceleration can become large.

The strong-field condition is

$$g \gg a_\star.$$

Then the term

$$\left(\frac{g}{a_\star}\right)^{123/50}$$

becomes large, and the suppression factor becomes small.

Thus,

$$\Sigma \rightarrow 0.$$

This is the strong-field limit.

This prevents the curvature-polarization response from running away. The nonlinear boost is active when needed, but it is muted when the field becomes too strong.

3 Early Galaxy Stability

A representative early epoch is

$$z \approx 14.$$

At this stage, CPTG allows rapid early structure formation because curvature polarization strengthens the effective gravitational response.

However, the same system must remain stable. Without suppression, the nonlinear response would continue increasing and could produce unphysical early galaxies.

The controlling factor is

$$\frac{123}{50}.$$

It allows galaxies to form rapidly, but it also limits the response once the system reaches high internal acceleration.

This gives early galaxies finite structure:

rapid formation $\rightarrow$ suppressed runaway $\rightarrow$ stable galaxy.

4 Luminosity Regulation

The suppression factor also limits luminosity runaway.

Curvature polarization can accelerate gas organization and star formation, but only up to the point where the local acceleration becomes large enough to activate the suppression term.

The result is a self-regulating loop:

curvature boost $\rightarrow$ rapid structure $\rightarrow$ high acceleration $\rightarrow$ suppression.

This allows early galaxies to be bright and mature without becoming unphysically massive or consuming their gas without limit.

5 Lattice-Lock Precision

The exponent is not arbitrary. It is

$$\frac{123}{50}.$$

In CPTG lattice form, this is a denominator-50 structural value.

A small shift changes the behavior of the governor. If the exponent is too low, suppression turns on too weakly and the early nonlinear response can become excessive. If the exponent is too high, suppression turns on too strongly and early galaxy formation is reduced.

Thus, the same suppression structure that stabilizes modern galaxy dynamics also controls the high-density early-universe regime.

6 Newtonian Recovery

The suppression factor also protects the Newtonian limit.

In strong-field regions,

$$g \gg a_\star,$$

the nonlinear CPTG contribution is suppressed. The effective response moves back toward the ordinary baryonic gravitational limit.

This is why CPTG can produce enhanced low-acceleration galaxy behavior while still avoiding excessive effects in dense regions.

7 Summary

The suppression factor is the governor of the CPTG galaxy equation.

Its exponent is

$$\frac{123}{50}.$$

It allows curvature polarization to support rapid galaxy formation, including early high-redshift systems, while preventing runaway collapse, excessive luminosity growth, and loss of the Newtonian limit.

In compact form:

$$\Sigma = \left[1 + \left(\frac{g}{a_\star} \right)^{123/50} \right]^{-11/50}.$$

This provides the throttle that keeps the theory stable across both early-universe and present-day galaxy regimes.